Supplementary Material

Adaptation-Aware Model Selection for Few-Shot Digital Twin Instantiation

This supplement provides feature definitions, model-bank specifications, calibration records, component sensitivities, and execution measurements supporting the main article. The primary method uses mean validation loss, a development-selected reference, and a separately calibrated relative margin. The evaluation is a retrospective, non-blind comparison; independence of the evaluation set is not established.

S1 Chronology and Feature Construction

The Bitbrains benchmark uses disjoint groups of 20 source, 20 development, 20 calibration, and 30 evaluation VMs. VM identities are listed in `data/data_roles.json`. Within every target task, records retain their native chronological order. An input at origin t contains records $t-95, \dots, t$; its target contains CPU values at $t+1, \dots, t+H$. Support windows precede the validation history. Table S1 gives the exact geometry for the four horizon/support conditions. Validation labels are fully observed at decision time t_0. The selected model predicts from origins $t_0, \dots, t_0 + B - 1$ with fixed weights as observations arrive. Forecast inputs may overlap the validation history, while forecast targets occur strictly after the decision time.

Table S1. Chronological task geometry (zero-based native-record indices).

H	K	Support records	Validation origins	V before mask	t_0	Forecast origins	B
1	10	0–105	201–1551	1351	1552	1552–2178	627
1	20	0–115	211–1556	1346	1557	1557–2181	625
4	10	0–108	204–1549	1346	1553	1553–2176	624
4	20	0–118	214–1555	1342	1559	1559–2179	621

Indices are relative to the target span's start. Each input ends at its origin and predicts the next H records. Validation labels end at or before t_0; forecast labels begin at $t_0 + 1$.

The support counts $K = 10$ and 20 specify labels used for parameter updates. The validation history supplies 1342–1351 complete candidate-comparison windows before target-validity masking. Few-shot parameter adaptation is therefore accompanied by a longer observed history for model selection. The common evaluation mask requires all H target CPU values to be finite and nonnegative.

The ten raw metrics are CPU cores, provisioned CPU capacity, CPU usage in MHz, CPU usage in percent, provisioned memory, used memory, disk-read throughput, disk-write throughput, received-network throughput, and transmitted-network throughput. For each raw metric, a binary validity indicator denotes a finite, nonnegative observation. Invalid values carry forward the latest valid value within the same span, with initial value zero.

The nine non-CPU-percent channels are transformed as

$$z_{t,j} = \frac{\log(1 + \widetilde{x}_{t,j})}{\max\{Q^{\mathrm{src}}_{0.99,j}, 10^{-6}\}}, \tag{S1}$$

where $Q^{\mathrm{src}}_{0.99,j}$ is the pooled 99th percentile of valid log-transformed observations from the first 86400 seconds of the 20 source VMs. CPU percentage and labels are divided by 100. Two additional channels are the trailing mean of up to five filled CPU values and the signed CPU difference from the preceding record. Their masks require validity of all raw CPU observations entering the corresponding computation; the first difference and its mask are zero. The final channel is the timestamp gap in seconds divided by 1200, initialized at zero. Together these form 12 value channels, 12 validity channels, and one gap channel. Recorded CPU values above 100% remain valid; transforms do not clip values. All operations use current or past records only. Exact fitted scales and per-task indices accompany the paper in `source_normalization.json` and `target_windows.csv`.

S2 Candidate Bank and Source Training

Architecture preselection uses 80 controlled synthetic cases (20 centers with four horizon/support settings) over 66 architecture configurations: 18 MLP, 36 TCN, and 12 GRU configurations. The check-best and validation-selected-positive counts for the retained configurations A1, A6, A13, A55, A56, and A57 are, respectively, $(1, 2)$, $(2, 1)$, $(2, 1)$, $(3, 2)$, $(2, 0)$, and $(6, 7)$. A configuration is retained when either count reaches two. These counts determine the architecture set; the reusable source checkpoints are trained on Bitbrains observations. The additional A57-PT initialization produces seven candidate records from the six declared configurations.

A1, A6, A13, A55, A56, and A57-PT use Adam with MSE, learning rate 10^{-3}, batch size 64, and 50 source epochs. Each source VM contributes 100 training windows per horizon, evenly indexed among valid windows in its first 86400 seconds. The optimizer uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and zero weight decay. A57-C1 pools the first ten valid windows per source VM and horizon. It uses ten source epochs of Adam/MSE followed by 180 source updates with a

Table S2. Candidate specifications and estimated inference complexity.

Record	Family	Layers	Width	Dropout	Params. ($H = 1/4$)	Ops. ($H = 1/4$)
A1	MLP	2	32	0.1	77921 / 78020	155712 / 155904
A6	MLP	3	32	0	78977 / 79076	157760 / 157952
A13	MLP	4	32	0.1	80033 / 80132	159808 / 160000
A55	GRU	1	16	0.1	2081 / 2132	377888 / 377984
A56	GRU	1	32	0	5697 / 5796	1050688 / 1050880
A57-C1	GRU	1	32	0.1	5697 / 5796	1050688 / 1050880
A57-PT	GRU	1	32	0.1	5697 / 5796	1050688 / 1050880

The declared dropout setting is inactive inside a one-layer GRU. A56, A57-C1, and A57-PT therefore share an executable structure but have different source-trained weights. Each record has separate checkpoints for the two horizons.

fresh Adam optimizer, Huber threshold 0.2, and sampled batches of 10 or 20 windows. Both stages use learning rate 10^{-3} and gradient-norm clipping at five. All seven records subsequently receive the same target adaptation: full-support SGD/MSE, 50 updates, learning rate 0.01, and gradient-norm clipping at one. Source-training differences enter through the initializations rather than the target update budget.

The one-layer GRU implementation does not apply inter-layer dropout. Consequently, A56 and both A57 records have the same executable GRU-32 structure; their distinct trained parameters remain separate bank entries. The parameter and operation estimates in Table S2 use input shape 96×25 and the horizon-specific output head. The analytic operation estimate counts a multiplication and addition as two operations: a dense transform contributes $2d_{\text{in}}d_{\text{out}}$, each GRU layer contributes $6L(dh + h^2)$, and a temporal convolution contributes $2Lkc_{\text{in}}c_{\text{out}}$. Output heads and residual projections are included; bias additions and elementwise activations are outside this counting convention.

S3 Reference Designation and Margin Calibration

Reference designation averages the assessment loss over four tasks within each development VM, then equally across the 20 development VMs. Candidates must satisfy every specified task/budget combination. Table S3 shows that A55 has the smallest eligible development loss. Exact development ties use maximum parameter count, maximum operation count, and record ID.

Table S3. Reference designation using the 20 development VMs.

Record	Dev MSE	Feasible for all	Max. params.	Max. ops.	Selected
A1	0.02987810	No	78020	155904	
A13	0.03577278	No	80132	160000	
A55	0.00403335	Yes	2132	377984	Yes
A56	0.00412156	Yes	5796	1050880	
A6	0.01455082	No	79076	157952	
A57-C1	0.01118221	Yes	5796	1050880	
A57-PT	0.00404386	Yes	5796	1050880	

Development MSE averages four task losses within each VM and then equally across VMs. All-task feasibility includes the tight budget.

Calibration uses the separate 20-VM pool and the same four horizon/support conditions. For a candidate margin, harm is a selected alternative's future MSE exceeding that of the adapted reference; retained-reference tasks contribute zero to the harm count. The denominator is all 80 calibration tasks. A margin is eligible when this empirical rate is at most 5%. Eligible margins are compared by mean selected-model MSE, with the larger margin resolving exact ties. The policy retains the reference if no margin qualifies or the smallest eligible loss does not improve on reference-only loss. Table S4 gives the primary calibration curve.

Table S4. Relative-margin calibration on 20 VMs (80 tasks).

τ	MSE (10^{-3})	Harm/80	Switch/80	Selected
0	2.852792300	4	64	
0.05	2.842945390	2	62	Yes
0.075	2.842991255	2	59	
0.1	2.842991255	2	59	
0.125	2.842991255	2	59	
0.15	2.863254059	2	57	
0.2	2.901653996	2	53	

The reference-only calibration MSE is 0.003037474. All seven margins satisfy the empirical 5% harm criterion; $\tau = 0.05$ minimizes selected-model MSE. Values use equal weighting across VMs.

At target selection, the primary policy uses the mean post-adaptation validation MSE and $\tau = 0.05$. Exact candidate-score ties use parameter count, operation count, architecture index, and candidate record ID, in that order. The

medium-budget policy is used at all three main deployment budgets. Calibration limits are empirical selection criteria; the observed evaluation harm rates for different source seeds are reported in Section S6.

S4 Error Metrics and Paired Uncertainty

For task i, let $\mathcal{W}_i$ contain rolling forecast origins with valid complete targets. For each origin, squared and absolute error are averaged over the H forecast components. Task MSE and MAE average these quantities over $\mathcal{W}_i$. Worst-10% averages the largest $\max\{1, \mathrm{round}(0.1|\mathcal{W}_i|)\}$ window-level squared errors within the task. The reported mean metrics average the corresponding task metrics over 120 tasks. Task-level CVaR90 instead averages the 12 largest task MSEs. It measures a different tail from Worst-10%.

The two reported relative improvements are

$$G_{\text{aggregate}} = 100\left(1 - \frac{\sum_i E_i^{\text{MSA}}}{\sum_i E_i^{\text{PT}}}\right), \tag{S2}$$

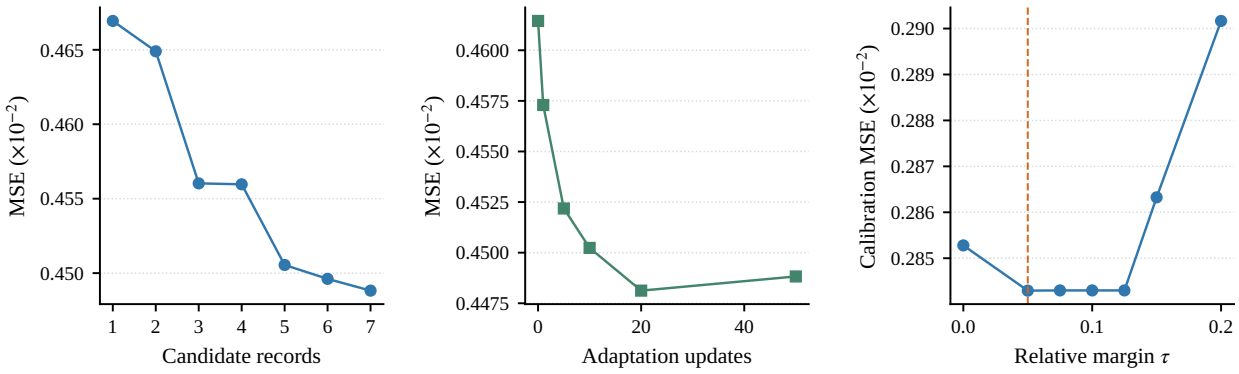
$$G_{\text{task}} = \frac{100}{120}\sum_i \left(1 - \frac{E_i^{\text{MSA}}}{E_i^{\text{PT}}}\right). \tag{S3}$$

They are 6.88% and 41.26%, respectively. Their difference reflects the weight assigned to tasks with different baseline error magnitudes. The abstract reports $G_{\text{aggregate}}$. The paired bootstrap resamples 30 VMs with replacement, keeps each sampled VM's four tasks together, and recomputes the statistic over 10000 draws (seed 20261026). Its percentile interval for $G_{\text{aggregate}}$ is [4.26%, 15.95%]. Subgroup intervals use the same VM resampling unit and the subgroup's task-relative statistic. These intervals describe variation across the sampled VMs in this workload trace.

All four main metrics use the same selected predictions. Relative to adapted A55, the 120 decisions comprise 84 beneficial alternatives, five harmful alternatives, and 31 reference retentions. A55 alone has CVaR90 0.036736 versus 0.036825 for the complete method, although its aggregate MSE is higher.

S5 Candidate Subsets, Update Counts, and Matched Optimizers

Figure S1 examines three components of model selection. The candidate subsets are nested in the order A55, A1, A6, A13, A56, A57-C1, and A57-PT. A55 remains the reference. Each subset and each update-count condition receives its own margin calibration. Tables S5 and S6 provide the exact values.



(a) Candidate subset (b) Update count (c) Margin calibration

Figure S1. Component sensitivities: (a) nested candidate subsets, (b) adaptation updates with seven candidates, and (c) the primary calibration-margin curve. Panels (a)–(b) report evaluation MSE; panel (c) reports calibration MSE.

The subset curve improves from 0.004669 with A55 alone to 0.004488 with seven records. The update curve is nonmonotonic: 20 updates yield 0.004481, slightly below 0.004488 at 50 updates, and agree with the 50-update candidate selection in 82.5% of tasks. Thus, the results do not imply that additional adaptation steps always improve selection.

The matched-optimizer comparisons in Table S7 use SGD/MSE, 50 updates, learning rate 0.01, and clipping at one for every adapted candidate. The A55 shortlist combines A55 with the first 11 non-A55 configurations in the fused-ranking shortlist, using their source-trained initializations. The A57-PT shortlist is formed analogously. Within each shortlist, calibrated reference-based selection and direct validation minimization share identical candidates, adaptation, and scores. The last two controls apply the matched optimizer to the fused-ranking and support-proxy NAS shortlists.

Direct validation minimization improves MSE from 0.004472 to 0.004429 in the A55 shortlist, with zero harmful selections for both policies. In the A57-PT shortlist it improves MSE from 0.004634 to 0.004614, while harmful selections increase from zero to one of 120 tasks. The relative-margin effect varies with the shortlist and reference.

Table S5. Nested candidate-subset sensitivity.

Records	τ	MSE
1	0	0.00466938
2	0	0.00464900
3	0.05	0.00456028
4	0.05	0.00455966
5	0.05	0.00450548
6	0.05	0.00449615
7	0.05	0.00448824

Table S6. Adaptation-update sensitivity.

Updates	τ	MSE	Agreement
0	0	0.00461449	60.83%
1	0	0.00457297	59.17%
5	0	0.00452183	55.00%
10	0.075	0.00450235	65.00%
20	0.05	0.00448124	82.50%
50	0.05	0.00448824	100.00%

Subsets follow the order stated in Section S5. Each subset/update condition has its own calibrated margin. Agreement compares candidate identities with those selected after 50 updates.

Table S7. Matched-optimizer controls (errors in 10^{-2} units).

Selection policy	τ	MSE	MAE	Worst-10%	CVaR90	Harm
A55 shortlist, calibrated	0.2	0.4472	1.9689	2.8063	3.6736	0.00%
A55 shortlist, validation minimum	0	0.4429	1.9353	2.7722	3.6736	0.00%
A57-PT shortlist, calibrated	0.2	0.4634	1.9798	2.8973	3.8389	0.00%
A57-PT shortlist, validation minimum	0	0.4614	1.9629	2.8837	3.8389	0.83%
Fused ranking, SGD50	0	0.7790	2.5407	4.6905	6.4468	–
Support proxy, SGD50	0	0.7306	2.4666	4.4837	6.0922	–

Each calibrated policy and its validation-minimum counterpart share the same task-specific shortlist and source checkpoints. Harm is relative to the named adapted reference; the last two controls have no reference-based policy.

S6 Source and Architecture-Preselection Sensitivity

Table S8 reports all source-pool and source-seed conditions. The source pools are nested; feature normalization remains fitted to the 20-VM source pool. Each trained bank receives a development-selected reference and a calibrated margin. The seed-2905 and seed-2906 banks have evaluation harm rates of 11.67% and 5.83%, exceeding the 5% empirical calibration criterion. Source training affects both prediction error and selection harm.

Table S8. Source-training sensitivity (errors in 10^{-2} units).

VMs / seed	Ref.	τ	MSE	MAE	Worst-10%	CVaR90	Harm
5 / 2904	A57-PT	0.125	0.9253	3.1955	3.5559	7.8912	2.50%
10 / 2904	A57-PT	0.075	0.5778	2.5996	3.2157	4.6889	3.33%
15 / 2904	A56	0	0.4658	2.2216	2.8728	3.7903	2.50%
20 / 2904	A55	0.05	0.4488	2.0543	2.8011	3.6825	4.17%
20 / 2905	A56	0	0.4728	2.1490	2.9780	3.8825	11.67%
20 / 2906	A57-PT	0	0.4518	2.0789	2.8611	3.6657	5.83%

All rows use the same evaluation VMs and 20-VM source-fitted normalization. References and margins are selected separately for each bank; harm is relative to that bank's reference.

Table S9 examines an additional bank-construction condition using Bitbrains development cases for architecture preselection. Three disjoint subsets contain seven, seven, and six VMs; their union is the full 20-VM development pool. Architecture evidence counts use these subsets or their union, after which retained configurations receive source training. Reference designation uses Dev20 and margin calibration uses Cal20. This diagnostic condition shares development data between architecture preselection and reference designation. The primary bank in Section S2 uses the separate synthetic preselection cases.

Table S9. Development-pool preselection sensitivity at the medium budget.

Pool	VMs	Tasks	$t = 1$	$t = 2$	$t = 3$	Ref.	τ	MSE (10^{-2})	Harm
Full Dev	20	80	28	22	8	A61	0	0.4464	10.00%
Subset 1	7	28	19	13	3	A61	0.05	0.4516	11.67%
Subset 2	7	28	11	8	2	A55	0.05	0.4444	10.00%
Subset 3	6	24	11	9	5	A57-PT	0	0.4575	5.00%

Threshold columns count retained architecture configurations. Downstream banks use $t = 2$ and include A57-PT as an additional reusable record. The three subsets partition the 20-VM development pool; the full-pool analysis overlaps all three.

The threshold-two architecture counts range from eight to 22 across pools. At the medium budget, MSE ranges from 0.004444 to 0.004575 and harm from 5.00% to 11.67%. Bank construction changes the feasible alternatives, the selected reference, and the calibrated policy. The full-pool condition overlaps its three subsets, so these four results are dependent sensitivity conditions. The accompanying CSV includes tight and loose budgets as well.

S7 Resource Constraints and Measured Computation

The primary bank has seven records at the medium and loose budgets and four at the tight budget. Under the tight budget, all seven records satisfy the operation limit, while the three MLP records exceed the parameter limit. The eight-record stress bank adds the source-trained temporal convolutional configuration A24. Its joint parameter and operation limits are 79554.5 and 1124496, respectively. Table S10 compares jointly enforced and single-active-limit conditions, each with its own calibrated margin.

Table S10. Resource-constraint diagnostics (MSE in 10^{-2} units).

Bank	Active limits	Feasible	τ	MSE	Harm
Q7 (tight)	Both	4	0.05	0.4532	2.50%
Q7 (tight)	Operations only	7	0.05	0.4488	4.17%
Q7 (tight)	Parameters only	4	0.05	0.4532	2.50%
Q8 (stress)	Both	6	0.05	0.4489	4.17%
Q8 (stress)	Operations only	7	0.05	0.4488	4.17%
Q8 (stress)	Parameters only	7	0.05	0.4428	4.17%

Each constraint condition has its own calibration. For the eight-record stress bank, parameter-only selection violates the operation limit in 16 tasks, and operation-only selection violates the parameter limit in two tasks.

With the eight-record bank, operation-only selection violates the joint parameter limit in two tasks; parameter-only selection violates the joint operation limit in 16 tasks. Both bounds have observable effects when they constrain different candidate records.

The hardware measurements use Windows 11, Python 3.13.5, PyTorch 2.13.0 with CUDA 12.6, and an NVIDIA GeForce RTX 3060 Laptop GPU. Table S11 reports the five executable structures at both horizons on CPU and GPU. Each setting uses an input tensor of shape $1 \times 96 \times 25$, 100 warmup calls, and five batches of 1000 timed calls. The CPU runs with one thread; GPU measurements synchronize around the timed forward pass. A deterministic zero-valued input standardizes this computation benchmark. Parameter and operation counts describe model complexity; latency also includes framework, device, and sequence-execution overhead.

Table S11. Single-window inference latency on the measured host (milliseconds).

Structure	H	Params.	Ops.	CPU median	CPU p95	GPU median	GPU p95
A1	1	77921	155712	0.0451	0.0533	0.1469	0.2065
A1	4	78020	155904	0.0452	0.0558	0.1322	0.2621
A6	1	78977	157760	0.0463	0.0551	0.1627	0.3414
A6	4	79076	157952	0.0462	0.0768	0.1485	0.2365
A13	1	80033	159808	0.0732	0.1105	0.2207	0.4686
A13	4	80132	160000	0.0731	0.0857	0.2025	0.2779
A55	1	2081	377888	1.9381	2.3867	5.9565	8.0760
A55	4	2132	377984	1.9492	2.4966	5.8619	7.2274
A56/A57	1	5697	1050688	1.9588	2.3125	5.9442	10.0755
A56/A57	4	5796	1050880	1.9688	2.4392	5.8404	7.2944

Each device/structure/horizon setting includes 100 warmup calls and 5000 timed calls. CPU inference uses one thread; GPU calls are synchronized. Latencies include framework and device overhead.

Target-pipeline timing uses calibration VM 1204, four horizon/support conditions, and three deployment budgets. Each method/condition has one warmup and five measured repetitions. The timed region includes checkpoint loading, model construction, search proxies where applicable, adaptation, validation, feasibility masking, and selection. Offline source training, prefix construction, and subsequent forecast evaluation are separate stages. The MSA-DTI implementation executes all seven candidate adaptations before budget masking, giving mean runtime 8.124 s versus 1.971 s for PT+FT. These measurements describe seven-candidate execution. The feasibility-first algorithm in the main article instead scales its adaptation work with the feasible subset. The file `pipeline_timing.csv` supplies the 96 method/condition summaries for the eight methods, including repetition counts, means, standard deviations, and ranges.

S8 Accompanying Data and Figure Reproduction

The LaTeX project includes plot-ready data and the manuscript figures. The data directory contains per-task errors, candidate identities, resource counts, and control results. The Python scripts reproduce experimental Figures 5–9 from the accompanying plot-ready data. The selection-rule code and manuscript sources are available in Release v1.3.0 at `https://github.com/w924871681/paper-code-DT-Twin/releases/tag/v1.3.0`. The accompanying numerical records reproduce the reported summary statistics. Raw workload traces are obtained from the Grid Workloads Archive at `https://atlarge-research.com/gwa-t-12/`.